VITALTWIN

SYSTEM ARCHITECTURE · PATIENT DIGITAL TWIN PROTOTYPE

Digital Twin Challenge 2026 · Team After Eclipse · SRM Institute of Science and Technology

Next.js 16 · React 19 · TypeScript 5.7 · client-side only · MIT

github.com/Akash-Selvaraj-R/VitalTwin

LAYERED DATA PATH

END-TO-END

1. **PATIENT**
   Person interacts with the prototype in a single browser session.
2. **ONBOARDING / PROFILE**
   Name, age, sex, height, weight, activity level — plus optional conditions, medications, allergies. Local-only.
3. **PERSONAL BASELINE**
   `computeBaseline(profile)` — age / sex / BMI / activity-aware heuristics derive one baseline value per signal.
4. **SIMULATED PHYSIOLOGICAL SIGNALS**
   8 synthetic signals advanced by deterministic hash noise toward the active scenario target. **No sensor, wearable or clinical feed.**
5. **VITAL TWIN ENGINE**
   `lib/engine.ts` — deviation analysis → sustained activation → hysteresis state machine → score → events.
6. **STATE / HEALTH SCORE**
   Twin state **STABLE · ATTENTION · RECOVERING** plus contextual score labelled ALIGNED / STEADY / SHIFTING / STRAINED.
7. **INTELLIGENCE EVENTS**
   Generated only for sustained, meaningful transitions — never per-tick noise.
8. **EXPLAINABILITY**
   Six-step causal chain attached to every event, in plain language.
9. **PATIENT ACTION / MONITORING**
   Safe, non-diagnostic next-step considerations; monitoring continues; history is captured.

⚠ SIMULATED / SYNTHETIC DATA

Every signal, history point and score in this prototype is procedurally generated on-device. Nothing is measured, transmitted, or clinically validated.

APPLICATION COMPONENT MAP

AS IMPLEMENTED

NEXT.JS APP (APP ROUTER) · REACT 19

`app/page.tsx — composition root` `app/layout.tsx — layout + analytics`

Boot splash, greeting, drawer + tour composition, focus management, keyboard handling (Escape closes layers).

CORE LOGIC (NO REACT, NO DOM)

`lib/engine.ts` `lib/use-vital-twin.ts` `lib/utils.ts`

engine.ts — pure deterministic simulation, baselines, deviation analysis, state machine, scoring, events, history.
use-vital-twin.ts — React binding: tick loop (cleaned up on unmount), persistence, scenarios, overlays, a11y.

REACT COMPONENTS

Onboarding · DigitalTwin + EcgStrip · MonitoringStrip · SignalCard ×6 · HealthScore · TwinSees · IntelligenceFeed · ExplainabilityDrawer · HistoryDrawer · AccountDrawers · TourOverlay · DemoControls

PERSISTENCE BOUNDARY — SINGLE-BROWSER PROTOTYPE STORAGE

`localStorage · vital-twin-profile` `localStorage · vital-twin-reduced`

Profile + reduced-motion preference only. Sanitized on read (malformed JSON and out-of-range values are repaired or cleared). **No backend, no cloud patient record, nothing leaves the device.**

TWIN STATE MODEL

HYSTERESIS-FILTERED

STABLE
No sustained deviation beyond personal tolerance.

ATTENTION
≥1 major or ≥3 sustained off-baseline signals.

RECOVERING
Deviations shrinking toward baseline; settles to STABLE.

These are the only three twin states. Demo **scenarios** (STABLE DAY · ELEVATED SIGNALS · RECOVERY) and **time controls** (+1 HOUR · +6 HOURS · PAUSE · RESET) are demonstration inputs that steer the simulation — they are deliberately distinct from the patient's twin state.

ENGINE & BOUNDARIES

DETERMINISTIC RULE / STATE ENGINE — NOT A TRAINED ML MODEL

lib/engine.ts — pure, framework-free, reproducible
`no Math.random · no inference runtime · no training data`

SIGNAL-TO-INTELLIGENCE PIPELINE

PER TICK

SIMULATED SIGNALS
8 signals advanced toward the active scenario target

▶

SIGNAL DEVIATION
current vs. personal baseline — % and direction

▶

TOLERANCE / THRESHOLDS
per-signal tolerance and major-deviation limits

▶

SUSTAINED ACTIVATION
must hold across consecutive ticks to count as evidence

▶

STATE ENGINE
hysteresis decision: STABLE / ATTENTION / RECOVERING

▶

HEALTH SCORE
weighted blend, shown with breakdown

▶

INTELLIGENCE EVENT
only on meaningful transitions + explanation payload

ENGINE CONSTANTS (FROM CODE)

REPRODUCIBLE

5	1200 ms	0.42	0.33	0.25
ACTIVATION_TICKS — ticks a deviation must hold before it counts as evidence	SIMULATION TICK — one signal update through the full pipeline	SCORE WEIGHT — baseline alignment	SCORE WEIGHT — recovery	SCORE WEIGHT — stability

Values describe this prototype's own simulation thresholds. They are **not** clinical limits, diagnostic criteria, or medical reference ranges.

DEMO CONTROLS vs. TWIN STATE

DEMO INPUTS

DEMO SCENARIO (input)	WHAT IT STEERS
STABLE DAY	Signals held near baseline — the Twin stays STABLE.
ELEVATED SIGNALS	Coherent multi-signal drift away from baseline → sustained deviation → ATTENTION.
RECOVERY	Targets set inside tolerance so the Twin visibly settles: ATTENTION → RECOVERING → STABLE.
+1 HOUR / +6 HOURS	Time fast-forward (4 sub-steps of 200 ms per simulated hour).
PAUSE / RESET	Freeze the simulation, or return it to the baseline state.

Scenario names are **not** twin states: ELEVATED SIGNALS and RECOVERY are demonstrator inputs; the patient-facing state is always STABLE, ATTENTION, or RECOVERING.

SAFETY BOUNDARY

NON-DIAGNOSTIC

- ✓ **SYNTHETIC / SIMULATED DATA**
  All signals, history and scores are generated on-device. No real patient data is used or required.
- ✓ **NON-DIAGNOSTIC LANGUAGE**
  “signal”, “pattern”, “elevated”, “outside your usual baseline”, “consider seeking medical advice”.
- ✓ **PROTOTYPE ONLY**
  Not a medical device. No clinical validation, regulatory approval, or diagnostic accuracy is claimed.
- ✓ **NO CLINICAL DEVICE INTEGRATION**
  Nothing is measured, connected, transmitted, or escalated.

NOT CURRENTLY IMPLEMENTED

OUT OF SCOPE

- ✗ **NO ABHA / ABDM INTEGRATION**
  No national health ID linkage.
- ✗ **NO EHR / HOSPITAL INTEGRATION**
  No clinical system is connected.
- ✗ **NO WEARABLE CONNECTION**
  No sensor or health-platform feed.
- ✗ **NO BACKEND / CLOUD PATIENT RECORD**
  Single-browser localStorage only.
- ✗ **NO REAL-TIME CLINICAL DATA**
  No real-time monitoring of any real patient.

PROTOTYPE BOUNDARY & FUTURE INTEGRATION

FUTURE SCOPE

The personal-baseline layer is deliberately replaceable: a learned, patient-specific baseline model could be substituted **without changing the surrounding UX or the explanation contract**. Possible future directions — **not implemented today** — include wearable integration, real longitudinal patient data, cloud sync, healthcare interoperability, ABHA/ABDM linkage, clinically validated predictive models, and hospital integration.